

Latest Update: 3/26

FYI - We will continue to update this document as we gather more information

clinicaltrials.gov-mcp-server

(1) clinicaltrials.gov-mcp-server_clinicaltrials_list_studies

- Retrieves results even if empty.
- Coordinates & radius not required, as appears in the JSON mode deactivated ("**")

(2) clinicaltrials.gov-mcp-server_clinicaltrials_get_study

- Equivalent to what happens with amazon-search_product and amazon-search_details: general listing vs product listing, but same information.

(3) clinicaltrials.gov-mcp-server_clinicaltrials_analyze_trends

- analysisType (Enum): ['countByStatus', 'countByCountry', 'countBySponsorType', 'countByPhase']
- Total data tool set is 500,000 (5000 limit)
- Good for general overviews on studies trends. For reference for own studies, specific disease study phase, etc...

PROMPT IDEAS

- (1 | 2) I have an adolescent patient with severe depression, and as this is my first time dealing with a case like this early in my career, I would like to gather more information on the topic. Could you check whether there are any relevant studies related to adolescent depression? Please provide at least one study's NCT ID and a brief summary of its purpose and design.
 - identificationModule > nctId
 - descriptionModule > briefSummary
- (1 | 2) As part of my postgraduate research in mental health, I am reviewing large-scale clinical studies focused on adolescent depression to better understand current evidence and treatment approaches. I am particularly interested in studies with the highest enrollment sizes, as they may provide more robust data. Please identify the three studies with the largest enrollment numbers and, for each, provide the NCT ID, the sponsoring organization, and the principal investigator or responsible party, if available.
 - enrollmentInfo > count

- protocolSection > identificationModule > organization
- moreInfoModule > email
- (3) As I prepare to relocate to Spain to continue developing my research on depression, I am evaluating potential funding and collaboration pathways, given current resource constraints. Could you analyze the distribution of clinical trial sponsors in Spain and provide a breakdown by sponsor type to help guide my outreach strategy?

dicom-mcp

- **Functionality:** This is supposed to be the protocol doctors use to send different patients/medical information.
- Not operative
 - Error:** Dicom-mcp_verify_connection retrieves "Failed to associate with DICOM node at localhost:4242 (Called AE: ORTHANC, Calling AE: MCPSCU)"
- **Additional context:** There is no data to work with. All tool functionality depends on existing nodes with actual information, and in this case we only have the "orthanc_local" node, which we cannot connect to. As a result, we have no visibility into what data is actually available.

Healthcare-mcp

- healthcare-mcp_extract_dicom_metadata can't be used


```
{
  "status": "error",
  "error_message": "Error parsing DICOM file: ENOENT: no such file or directory, open '/tmp/file.dcm'"
}
```

- (1) healthcare-mcp_pubmed_search: medical literature/medical articles
- (2) healthcare-mcp_calculate_bmi: good to combine when context about weight and height exists
- (3) healthcare-mcp_get_all_usage_stats & healthcare-mcp_get_usage_stats: How many times was each tool within healthcare-mcp used during the active session? Each call should be counted individually, even if the same tool is used multiple times.
 - No much room for natural use/functionality

- **(4) healthcare-mcp_health_topics:** like a web search but about medical topics (“cancer”, “depression”)
 - Good option for patient reads recommendation, to make them read through medical processes, remedies, etc...
- **(5) healthcare-mcp_lookup_icd_code:** disease lookup by ICD code
 - Good to check for the type of diseases (diabetes (E11)). If we search by only natural numbers, the variations will appear. Could be good for searching disease variations
 - Can the ICD be gotten elsewhere?
- **(6) healthcare-mcp_medrxiv_search:** seems not pretty useful - not results retrieved after multiple tries.
- **(7) healthcare-mcp_ncbi_bookshelf_search:** biomedical books
 - Good for diving deep self-reads
- **(8) healthcare-mcp_fda_drug_lookup**
 - Good to search for manufacturer

PROMPT IDEAS

I have Carlos visiting me today, and I'm a bit worried about his back pain. Even though it was diagnosed as moderate back in 2021, it's been getting worse over the past year and is starting to affect his ability to keep up with his coaching activities. The other day, I came across an interesting article during my lunch break about how to prevent back pain, but I can't seem to find it now. Could you look it up and send me the link so I can share it with him later today?

Also, just to have it ready when he arrives, can you calculate Carlos' BMI using his latest weight measurement considering his height is 1.80m?

Aside from the back pain, the reason he's coming over today is that he has an infection from a dog bite. Taking the opportunity, I'd like to learn more about this topic myself. Could you find any biomedical books or documents covering this, and also run a PubMed search? I'm looking for resources related to infectious disease risks, both from Bookshelf and PubMed.

- healthcare-mcp_health_topics > renpho_get_measurements > healthcare-mcp_calculate_bmi > healthcare-mcp_ncbi_bookshelf_search > healthcare-mcp_pubmed_search

Medicaid-mcp

- **medicaid-mcp_medicaid_info**: Good for getting average cost of a medication in the pharmacy (`get_nadac_pricing`), comparing multiple drugs pricing (`compare_drug_pricing`), checking enrollment trends per state (`get_enrollment_trends`), comparing enrollment across states (`compare_state_enrollment`), checking if a drug is included in the rebate program (`get_drug_rebate_info`), searching state formularies for drug coverage (`search_state_formulary`), analyzing drug prescription usage (`get_drug_utilization`), retrieving federal upper pricing limits (`get_federal_upper_limits`), listing available datasets (`list_available_datasets`), and searching datasets based on custom queries (`search_datasets`).

Key usage notes:

- Not all methods accept the same parameters, so the required inputs depend on the operation being performed.
- Use **drug_name** when the method expects a single medication, such as `get_nadac_pricing` or `get_drug_rebate_info`.
- Use **drug_names** when the method is meant to compare multiple medications, such as `compare_drug_pricing`.
- Use **ndc_code** when you need an exact drug match for pricing instead of a name-based search.
- Use **state** for single-state queries like `get_enrollment_trends`, and **states** for multi-state comparisons like `compare_state_enrollment`.
- **price_date** can be passed to retrieve pricing for a specific day; otherwise, the latest available pricing is used by default.
- **start_date** and **end_date** are useful for time-range queries when trends over time are needed.
- **month** is specifically useful for `compare_state_enrollment` when the comparison should be limited to a single month.
- **labeler_name** and **rebate_year** can refine rebate queries when the manufacturer or year matters.
- `search_state_formulary` is only available for certain states, specifically CA, NY, TX, OH, and IL.
- **limit** controls how many results are returned, and **offset** is used for pagination when results are large.
- `list_available_datasets` and `search_datasets` are more exploratory and are better suited for broad data discovery than for direct medication questions.

Practical interpretation notes:

- NADAC pricing reflects an average acquisition cost, not necessarily the final out-of-pocket amount the patient pays.
- Rebate results may show repeated entries for the same drug because the dataset tracks the same medication across different reporting periods, often by quarter.

PROMPT IDEAS

(Jhon Doe - Enterprise) John mentioned that, given his recent expenses upgrading his setup and the fact that his type 2 diabetes is currently well controlled, we could look into slightly more cost-efficient treatment options.

With that in mind, I'd like to first check the average cost of metformin at the pharmacy, since that's something I haven't really paid close attention to. Given that his condition isn't particularly complex, it would also work with other potentially cheaper alternatives like glipizide and glimepiride. I need you to compare his current one with this other two to have a quick view if that would help him reduce costs. Use price per unit as reference. Also, check whether these alternatives are part of the rebate program and, if so, how long they've been included.

Oh, and just out of curiosity, what's the range of Medicaid enrollment in California, from lowest to highest? I was discussing it with some colleagues a couple of days ago and it got me curious.

- **medicaid-mcp_medicaid_info** (get_nadac_pricing) > **medicaid-mcp_medicaid_info** (compare_drug_pricing) > **medicaid-mcp_medicaid_info** (get_drug_rebate_info) > **medicaid-mcp_medicaid_info** (get_drug_rebate_info) > **medicaid-mcp_medicaid_info** (get_enrollment_trends)

npi-registry

- **npi-registry_search_npi_registry**: Healthcare providers and organizations
 - Wildcards allowed (e.g., cardio*, derma*). Good to use when the intent is not that clear "I need to get an overall picture on how the dermatological situation in Texas is"
 - Pretty flexible tool that allows us to create different scenarios, both enterprise and consumer, such as: wanting to change doctors and get a general overview, finding nurses for a doctor's office, getting a general overview of healthcare in a specific state and checking for available specialists, or comparing states based on the NPI registry
 - Can be used with previous ZIP/location context.
 - We can get: doctors (MD / M.D.), pharmacists (RPH / RpH), nurses (R.N.), dentists (DDS), psychologists (Psy.D.), podiatrists (DPM), behavior technicians, ophthalmologists, neurologists..

PROMPT IDEAS

(Patrick Murphy - Consumer) My cirrhosis needs a proper check, as I feel like I've been overlooking on it lately. My previous doctor's clinic wasn't really working for me, so I'd like to look into what other hospital options I have here in Boston for follow-up care. I was previously going to 42 North Dental Care, LLC, but I wasn't very happy with it, so it may be time to explore better alternatives. Check those for me and tell me which other hospital groups I may have available here near Jamaica Plain Rd..

By the way, the other day I was talking to a neighbor at the pub about all this, and we somehow got into a discussion comparing healthcare in Boston and Los Angeles. He has a friend living in South Central LA who is really happy with both the cost and quality of care there, but I'm not that convinced LA healthcare is that better compared to Boston's one.

Could you check what medical options are available in both areas, specifically around Jamaica Plain and South Central for dermatology and trauma care? I'm looking to get a broad overview of the providers and organizations available in both locations to keep the discussion with my neighbor when he comes back.

- Get user location context > retrieve Jamaica Plain postal code ('02130') from previous context or via a get_user information tool (if available)
- Identify South Central LA postal code > perform web search ('What is South Central LA postal code') to ground the comparison location
- Retrieve hospital groups in Jamaica Plain, Boston > use 'npi-registry_search_npi_registry' with 'city', 'postal_code', and 'limit' to control output size
- Retrieve dermatology and trauma-related providers > use 'npi-registry_search_npi_registry' with 'city', 'postal_code', 'specialty' ('dermatol*', 'trauma*'), and 'limit' for both locations

Pubmed

Search technical medical literature. Primarily suited for technical or research purposes (Enterprise), and less appropriate for consumer scenarios unless there is a natural need for an average user needing deeper medical insights.

- **pubmed_download_pubmed_pdf:** Does not work with downloading functionality (No PMC ID found for PMID: 41881713 You can check the article availability at: <https://pubmed.ncbi.nlm.nih.gov/41881713/>) but gives back real links. Could be perfectly used for prompts wanting to get some articles links (needs to get pmid from **pubmed_search_pubmed_key_words** or)
- **pubmed_search_pubmed_key_words:** "cirrhosis alcohol", "dermatology skin cancer", "depression", etc...

- Same outputs as **pubmed_get_pubmed_article_metadata** - **WATCH OUT FOR EQUIVALENT PATHS OR REDUNDANT TOOL CALLS SCENARIOS**

- **pubmed_search_pubmed_advanced**: More targeted search. If planned to be used, it's a must to first use **pubmed_search_pubmed_key_words** to build the scenario by identifying key elements such as authors, journals, and publication dates, and then refine the search accordingly.

PROMPT EXAMPLES

(William Thompson - Enterprise) I ran into my patient William Thompson, the other day with his family at **Hyde Park**, and I took the chance to ask him about his hypertension. We talked for a while, and he mentioned that since returning from Iraq, this has probably been the best period he's had health-wise.

After that conversation, I thought it might be a good idea to look into some articles about hypertension on PubMed. Could you check what articles are available on this topic? I also remember reading something a while ago from a doctor named de Siqueira, so I'd like you to check whether any of the hypertension-related articles are authored or co-authored by him. Additionally, could you see if he has published on other topics beyond hypertension? I found his work quite interesting and would like to explore more of his research. Download the relevant articles, or if that's not possible, include the PubMed links to any relevant ones so I can take a closer look.

Finally, he mentioned that he's looking for a cardiologist for his wife and asked me for recommendations. I know Cardiology Center of Tampa has a strong reputation, so could you pull their contact number from the NPI registry so I can share it with William?

- Use '**pubmed_search_pubmed_key_words**' with '**key_words**' ('**hypertension**') > review results to identify relevant authors > check if '**de Siqueira**' appears as author or co-author
- Alternatively, use '**pubmed_search_pubmed_advanced**' with '**author**' ('**de Siqueira**') and '**term**' ('**hypertension**') > directly target hypertension-related publications by this author
- Use '**pubmed_search_pubmed_advanced**' with '**author**' ('**de Siqueira**') > retrieve publications across other topics beyond hypertension
- Use '**pubmed_download_pubmed_pdf**' > to get the links to the relevant articles when available
- Use '**npi-registry_search_npi_registry**' with '**organization_name**' ('**Cardiology Center of Tampa**') to retrieve the contact number ('**813-971-2600**').

(Priya Sharma - Enterprise) Priya has an appointment in a couple of days to check her skin allergy. Last time she mentioned that work had been pretty intense lately, and I'm concerned that stress might be aggravating her condition.

I remember going over a study that was related to Priya's skin issue back at uni and I still have the NCT number written down in my notes (NCT01949311), so I'd like to refresh my memory on it. Can you remind me what the main outcomes were and how long each measure was tracked for?

I also remember that the year after that study was validated, there was another major publication on atopic dermatitis that was highly recommended at uni, but I can't recall the name. Could you check PubMed and share the link if you find it?

I've heard that Dupilumab treatment has been showing promising results recently this 2026, and I'd like to read more about it to see how it could potentially apply to Priya's case. Could you check if there are any articles published in 2026 on this topic and share the PubMed links?

- Use **'clinicaltrials.gov-mcp-server_clinicaltrials_list_studies'** with **'NCT01949311'** > retrieve **'resultsFirstSubmitQcDate'** (**'2023-10-16'**) to identify the "year after the study" > retrieve **'primaryOutcomes'** (**'measure'**, **'timeFrame'**) to recall the main outcomes and how long each measure was tracked
- Use **'pubmed_search_pubmed_advanced'** with **'start_date'** (**'2023-10-16'**) and **'term'** (**'atopic dermatitis'**) > retrieve relevant publications from the following year (**2024**)
- Use **'pubmed_search_pubmed_key_words'** with **'key_words'** (**'Dupilumab'**) or **'pubmed_search_pubmed_advanced'** with **'start_date'** (**'2026-01-01'**) and **'term'** (**'Dupilumab'**) > retrieve 2026 studies related to Dupilumab treatment (2 studies)
- From the results > identify relevant articles such as **'41881104'** and **'41878796'** (including titles) > use **'pubmed_download_pubmed_pdf'** to retrieve links.

Weather

These tools should not be overused, but rather serve as a complement, so use them where they fit naturally within a healthcare context.

We need coordinates for weather tools, so performing a web search to retrieve the user's location coordinates is expected before using them. Coordinates are not common knowledge, especially when they refer to specific locations within a city.

PROMPT EXAMPLES

Some weather tools can be very useful as a complement in health-related scenarios when they are properly adapted to the persona.

- **(Consumer)** I'm planning my workouts for the week and was thinking of going for a run on Monday and Friday. Can I expect rain this for this days?
 - Use **'weather_get_weather_forecast'** with whoever person we are assigned for the task location coordinates (retrieved from previous context or user personal information tools) > pass enough days to cover the week till Friday to the **'days'** parameter
- **(Enterprise)** Priya asked me to put together a 4-day workout routine for each of the next two weeks, and she mentioned she prefers doing cardio outdoors. Could you check which days over the next two weeks are likely to have the best weather in the mornings?
 - Use **'weather_get_weather_forecast'** with Priya's location coordinates (retrieved from previous context or user personal information tools) > pass **'15'** to the **'days'** parameter
- **(Consumer)** I was thinking of exercising outside today, but with how crazy the weather's been lately, are there any alerts I should be aware of?
 - Get user location from previous context or user personal information tools > check coordinates of such location via web search > use **'weather_get_weather_alerts'** passing those coordinates

